

Reference-relative believability: identity, realism, and interactivity in photoreal human avatars

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Abstract

This paper comes from building avatar systems, not from trying to name a narrow talking-head effect. The goal was a reusable engine where character, voice, language model, and persona become a living interface: support agent, assistant, virtual friend, presenter, virtual model, or full-body synthetic human. Most useful identities are authored characters, not copies of one real person. That does not make them free of reference. A known real person has an external reference in viewer memory. A persistent synthetic avatar has an internal reference in canonical face, body, voice, provenance, policy, and continuity state. Both can drift. The central claim is reference-relative believability: avatar quality is judged against the reference available for the task, not against pixels alone. The design consequence is an identity-contract router. Use the high-fidelity, gated path when the output must be indistinguishable under close human inspection. Use the low-latency path when live presence matters more than full-body photographic proof. Known-person likeness is one strict subcase, not the whole thesis.

1. Introduction

Photoreal avatar demos have become convincing in narrow crops. A single image and a few seconds of audio can produce a talking head that blinks, moves its mouth, and answers questions with a face attached. Those demos are useful, but they are also misleading. The product I kept trying to build was not just a talking head. It was a reusable avatar engine: character + voice + language model + persona -> living interface. The same engine should support a customer-service agent, a personal assistant, a synthetic presenter, a virtual friend, a virtual modeling roster, and eventually full-body synthetic humans that look like real people under high-definition inspection.

That broader target changes the problem. Most of the useful avatars are not supposed to be a known celebrity, a public figure, or the owner of the system. They are made-up characters. They still need to stay themselves across sessions, outfits, poses, camera angles, lighting, motion, and product surfaces. A modeling agency does not need a copy of one person; it needs reusable virtual talent. An assistant does not need to be a real employee; it needs a stable, believable presence. A content studio does not need to impersonate someone; it needs a synthetic human that can be reused without looking like a different person every time.

I spent several months, and close to a year if I include the earlier digital-double work, chasing that target through the wrong abstraction. Early systems treated support agents, assistants, presenters, full-body avatars, and known-person likeness as variants of the same renderer problem. They were not. Full 3D reconstruction ran into capture, license, and fidelity walls. Blendshape bridges and procedural idle motion looked like math. Single-image reenactment held some skin detail but drifted head shape and expression. Mouth-only lip sync preserved most of a real face but still failed around the mouth, teeth, and temporal boundary. Full-body generation made the failure broader: hands, eyes, hair, skin texture, face pixel size, garment fit, and pose consistency all became visible. A portrait that looked acceptable became weak once the whole body, wardrobe, camera, and motion had to survive.

The useful lesson was not simply that one method failed. It was that the reference changed the bar. A known real person is judged against memory. A persistent synthetic human is judged against its canonical identity state and against the viewer's category model of real humans. A throwaway generic face is judged against a much looser bar. Those are not the same product requirement.

This paper names that distinction reference-relative believability. The question is not only "does the frame look real?" It is "real relative to which reference?" For a known person, the reference is external and held by the viewer. For a persistent synthetic character, the reference is internal and maintained by the system. For full-body indistinguishability, the reference is the ordinary human body under close inspection. The system has to know which reference it is trying to satisfy before it chooses a renderer, a latency target, a safety policy, or an evaluation gate.

2. The tension nobody resolves

Avatar products usually collapse three requirements into one word: realism. In practice they are separate.

The first requirement is **human realism**: skin, hair, eyes, hands, body proportions, clothing, camera behavior, motion, and lighting must look like a real scene. The second is **identity continuity**: the same character must remain the same character across outputs. The third is **live interactivity**: the avatar must respond to unplanned speech fast enough to feel present. A serious product cannot treat these as one scalar score.

Full-body work makes the conflict obvious. Portrait crops hide many failures. They keep the face large, remove hands, avoid most garment physics, and constrain the pose. Full-body images expose the system. Face pixels shrink below reliable identity-measurement thresholds. Hands enter the frame. Hair, fabric, knees, shoes, posture, and body proportions matter. A model that passes a headshot test can fail a 96-image full-body variation grid because the prompt asks for different poses, outfits, lighting, and camera positions. That was the most important build lesson: solving framing revealed the detail problems that cropped demos had been hiding.

Live interaction creates a different pressure. You cannot pre-record the answer to a question you have not been asked. A conversational avatar has to render or drive speech-linked motion from text or audio. Today that usually means a learned talking-head model, a portrait warper, an audio-to-video model, or a clip library with lip-sync layered on top. These systems can be good enough for presence. They are not automatically good enough for high-definition full-body inspection.

The hard product problem is therefore not "can a model generate a face?" It is deciding which bar the output must clear. A full-body virtual model for a clothing shoot needs slower, gated, high-resolution generation and continuity checks. A live support assistant

may accept a narrower head-and-shoulders render if it answers quickly and honestly. A known-person likeness has an even stricter external reference because viewers already know the source.

3. The reference-relative principle

Here is the claim in plain form. A synthesized avatar is judged against the strongest reference available for the task.

For a known real person, the reference is the viewer's stored model of that person's appearance and motion. The viewer is not evaluating a generic human image. They are comparing against memory: jaw width, asymmetry, expression timing, eye behavior, skin and beard texture, the way the mouth moves when thinking. Familiar-face research already shows that known faces are processed differently from unfamiliar faces [4-8]. Recent deepfake-detection work makes the same point from the other direction: a 2025 systematic review finds that viewers spot manipulations more reliably when they already know the person's face [16]. In synthesis, that advantage becomes a stricter rejection mechanism.

For a persistent synthetic character, the reference is not viewer memory of a real person. It is the character's canonical identity state. The system defines the face, body, age band, voice, allowed styling range, provenance, and continuity rules. From that point on, the character is no longer a disposable prompt result. It is a roster identity. If the nose changes, the body proportions drift, the face ages across shoots, or the hands and skin fail close inspection, the output is wrong relative to the identity contract even if no real person was copied.

For a one-off synthetic stranger, the reference is looser: the image must belong to the category "real human" for the context. That is still not easy. Dead eyes, waxy skin, broken fingers, incoherent lighting, uncanny mouth motion, and temporal jitter all fail. What the one-off stranger escapes is only the tighter comparison against a known person or a canonical roster identity.

I use **ground-truth drift** for deviation from the applicable reference. The source of ground truth can be external memory, internal canonical identity, or a real-human category bar. The mistake in the early architecture was treating made-up characters as if they had no ground truth. That is true only before the character is committed. Once a synthetic identity is saved and reused, it has a reference, and the system has to defend it.

4. The split that survived the build

The split is not "real person versus fake person." That was the wrong shortcut. The split is between render contracts.

The **high-fidelity identity path** covers any avatar that must survive close visual inspection: a known real person, a fixed synthetic roster identity, a virtual model for fashion shoots, or a high-value presenter where screenshots will be inspected. This path is allowed to be slower. It needs canonical identity state, full-body framing, high-resolution generation, face and body continuity checks, skin and hand evaluation, garment realism review, provenance, and human review when the gate is uncertain. If the identity is a real person, the path should preserve real pixels or use extremely constrained edits. If the identity is synthetic, synthesis is allowed, but only under continuity gates. "Made up" does not mean "anything goes."

The **live interaction path** covers support agents, assistants, virtual friends, and other conversational surfaces where responsiveness is part of the product. It can use audio-driven talking heads, live portrait animation, pre-rendered idle clips, or streaming video chunks. Its job is presence, timing, voice, memory, and turn-taking. It still needs safety, disclosure, and continuity, but it should not pretend to be a full-body, high-definition proof surface unless it actually clears that bar.

Known-person likeness sits inside the first path as the strictest subcase. The earlier mistake was letting that subcase dominate the whole paper. The broader product target is harder and more useful: persistent synthetic humans that remain stable enough for modeling, assistant, companion, and content-creation workflows while looking indistinguishable from real people at the level users actually inspect.

5. What the principle predicts

The claim is useful only if it predicts something beyond "models will improve."
Reference-relative believability predicts different failure curves for different references.

Familiarity gradient. For the same synthesis pipeline and visible artifact level, ratings should get worse as the viewer knows the depicted real person better. The ordering should be easier for a stranger, harder for a public figure, harder again for a close relation, and hardest for the self.

Canonical-continuity gradient. Persistent synthetic characters should become stricter over time. The first exposure may be judged mostly against the human category. After a character is saved, reused, named, and seen across sessions, viewers and the system both acquire a reference. A render that would pass as a one-off stranger can fail as "not the same avatar."

Full-body exposure effect. Headshot demos should overstate quality. Moving to full-body framing should reveal failures in hands, hair, body proportions, garment interaction, face pixel size, pose, and lighting. If the model only passes when the face is large and the body is hidden, it has not cleared the full avatar bar.

Identity-swap rescue, with limits. A renderer that fails on a known real person may rate higher when the same motion and rendering method are applied to a new synthetic identity. That rescue should shrink once the synthetic identity has a canonical state and repeated exposure.

Latency-fidelity trade. A low-latency path may score well for presence and poorly for close inspection. A high-fidelity path may score well for screenshots and poorly for live turn-taking. Treating those as one score should produce confused product decisions.

The first study does not need a new model. Use one renderer and produce paired clips or images across four conditions: known real person, unfamiliar real person, newly authored synthetic identity, and persistent synthetic identity with a prior reference set. Hold duration, framing, voice quality, and prompt content as constant as possible. Separately rate visible artifact severity. The key question is whether reference condition predicts rejection after artifact severity is controlled. If it does not, the theory is wrong or incomplete.

6. Where this sits

I am not proposing another renderer here. The renderer work exposed a systems boundary.

The uncanny valley, from Mori onward [1], gives the broad discomfort shape when an entity becomes nearly human but not fully convincing. Later accounts emphasize motion, category conflict, and prediction error [2, 3]. Recent avatar studies sharpen this further: discomfort tracks the consistency of realism cues and the interaction context rather than raw fidelity, and the effect of appearance realism is moderated by task and setting [14,

15]. Reference-relative believability fits that literature, but shifts the unit of analysis. The relevant object is not just the stimulus. It is the stimulus in relation to a viewer and to the identity contract the system claims to preserve.

The face-perception literature explains the known-person subcase. Familiar and self faces are processed differently from unfamiliar faces [4-8]. That is why a near miss on a known person is rejected so quickly. But persistent synthetic identities add another case: once the system defines a canonical identity and repeats it across contexts, it creates its own reference.

Talking-head and portrait-animation systems provide the live surface [9-12]. Unified audio-to-video systems suggest that live avatar quality will keep improving [13]. Full-body generation and virtual-model workflows add a different surface: not just the face in motion, but the whole human body, wardrobe, camera, and scene. Better generators reduce drift. They do not remove the need to state which reference the output must satisfy.

7. The architecture this implies

The architecture I landed on keeps the conversational stack shared and makes the render contract explicit.

The shared stack includes speech recognition, language model routing, voice selection, memory, tools, transport, and policy controls. It can serve many products. The render layer sits behind a fixed interface, but it is not selected casually. An identity contract router decides whether a request belongs on the high-fidelity identity path or the live interaction path.

REFERENCE-RELATIVE BELIEVABILITY

Reference-aware Avatar Engine

The renderer is selected by identity contract, inspection bar, and latency target.

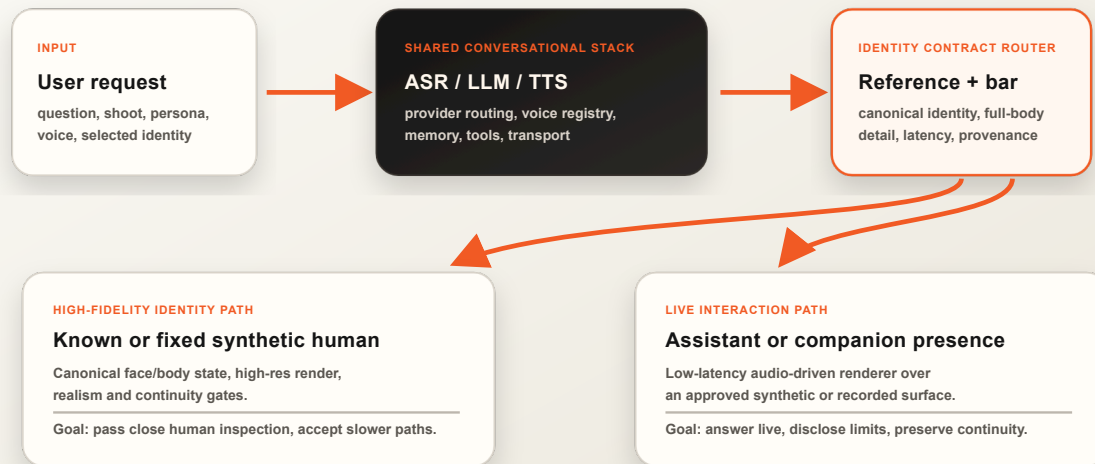


Figure 1. Reference-aware Avatar Engine architecture. The conversational stack is shared. The renderer is selected by identity contract, inspection bar, latency target, and provenance requirement.

The high-fidelity identity path treats a character as more than a face prompt. It carries canonical face state, body state, allowed styling range, identity references, evaluation thresholds, provenance, and safety policy. Outputs are gated for realism and continuity: face, skin, hands, body, garment interaction, camera coherence, and identity drift. This is the path for virtual modeling, high-value presenters, known-person likeness, and any artifact expected to survive zoomed-in review by a human viewer.

The live interaction path treats presence and timing as first-class requirements. It can use a narrower crop, a talking-head renderer, pre-rendered idle clips, a browser stream, or a real-time provider. It still uses the same character registry and policy layer, but it should be labeled as a live interaction surface rather than a proof of full-body indistinguishability.

In my current implementation, the boundary is concrete. The private Avatar Engine repository defines character registries, a renderer protocol, and user-choice registries for character, voice, and language model. A local loop connects speech recognition, an Ollama language model, Piper voice synthesis, and a Ditto talking-head renderer. TensorRT made the Ditto path fast enough for the real-time tier. A resident worker keeps the renderer warm, and a streaming path can render answer chunks as they complete. In parallel, separate high-fidelity identity-planning work defines the other side: identity

creation, roster state, full-body variation grids, realism gates, continuity services, provenance, and human review.

The unfinished parts are also part of the result. Full-duplex barge-in, browser transport, idle behavior, latency polish, full-body video, and garment-level production remain engineering work. The theory does not make those easy. It says they should not be hidden behind one generic "photoreal" label.

8. Limitations and safety

I have not measured the full curve yet. I have a build history, falsifiable predictions, and a working architecture boundary, not a completed human-subjects study. The curve may vary by culture, screen size, task, exposure duration, artifact type, and whether the user is judging utility, comfort, trust, or visual truth.

The reference can also move. A synthetic assistant that starts as a new character may become familiar after repeated use. A virtual model roster may start with an authored canonical state and then tighten as clients approve specific shoots. A model upgrade may improve general realism while breaking continuity. The architecture has to treat identity state as durable product data, not as prompt history.

This is not an argument for irresponsible synthetic humans. It is the opposite. Synthetic identities should be created with consent-safe source rules, age safeguards, similarity checks, disclosure, provenance (for example, signed content credentials [17]), and audit logs. Known-person likeness needs explicit consent and stricter renderer constraints. Persistent synthetic identity needs continuity gates and policy controls because realism increases the practical risk of misuse. A system that cannot say where an identity came from, what it is allowed to do, and which renderer may touch it is not ready.

9. Conclusion

The first version of this argument was too narrow. It made the known-person case the center. The bigger and more useful thesis is that photoreal avatars are reference-relative systems. A known person is judged against memory. A persistent synthetic human is judged against canonical identity state and the real-human category. A live assistant is judged against presence, timing, disclosure, and continuity. A full-body virtual model is

judged against high-definition human detail: face, skin, eyes, hair, hands, body, garment, camera, and motion.

The practical rule is simple. Do not ask one renderer to satisfy every reference. Route by identity contract. Put full-body, close-inspection avatars on a gated high-fidelity path. Put conversational presence on a live interaction path. Treat made-up characters as real product objects once they are saved. That is not a retreat from the original avatar-engine vision. It is the version that survived contact with the work.

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